ET AI HACKATHON 2026 · PROBLEM STATEMENT 8

PlantBrain

The Unified Asset & Operations Brain

AI for Industrial Knowledge Intelligence — one intelligence layer over every plant document, validated on 8,076 real maintenance work orders and official Indian regulatory standards.

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July 2026

1. Executive Summary

Indian heavy industry does not have a data problem — it has a memory problem. Professionals in asset-intensive industries spend 35% of working hours searching for information that already exists (McKinsey, 2024); the average large Indian plant runs 7–12 disconnected document systems (NASSCOM-EY); fragmented equipment history contributes to 18–22% of unplanned downtime (BIS Research); and 25% of India's experienced industrial engineers retire within the decade, taking undocumented knowledge with them.

PlantBrain is an AI platform that ingests every operational document a plant has — CMMS work orders, inspection registers, incident reports, SOPs, and regulatory standards — into one queryable, continuously watching intelligence layer. Four agents operate on that layer: an Expert Copilot that answers any operational question with citations; a Compliance Radar that continuously joins statutory requirements against inspection evidence; Maintenance Intelligence that detects repeat-failure patterns; and a Risk Echo engine — our core innovation — that raises an alert when the conditions that preceded a past incident become active again.

Unlike typical document chatbots, every claim is measured: entity extraction scores **94.8% token accuracy (0.909 macro-F1)** on held-out, expert double-annotated real work orders from the published MaintIE benchmark, and the working prototype ingests 8,076 real field work orders plus the official OISD-STD-116 standard and the Factories Act 1948.

2. The Problem

35%	7–12	18–22%	25%
of engineer hours spent hunting for information (McKinsey 2024)	disconnected document systems per large plant (NASSCOM-EY)	of unplanned downtime tied to fragmented history (BIS Research)	of experienced engineers retiring this decade (Problem Statement 8)

Knowledge fragmentation is not a file-management inconvenience. The incident record of Indian heavy industry shows the same pattern repeatedly: the information needed to prevent a failure existed — in a maintenance log, a lab spreadsheet, a contractor PDF — but nobody could see it at the moment of decision. It is a safety problem, a statutory compliance problem, and an operational efficiency problem, and it compounds every year.

3. The Solution: Four Agents on One Knowledge Layer

- **Expert Copilot** — ask the plant anything. GraphRAG-lite retrieval over 8,540 unified records; answers synthesized by an LLM (Groq, llama-3.3-70b) strictly constrained to retrieved context, with an automatic extractive fallback if the API is unavailable. Every claim cites an openable source record; knowledge-graph-sourced citations are visibly badged.
- **Risk Echo Engine** — cross-references every historical incident's assets against currently live risk signals (overdue statutory evidence, repeat-failure patterns) and raises cited warnings *before the repeat*. This is the layer that turns an incident archive into a prevention system.

- **Compliance Radar** — a continuous join of the regulatory library (OISD, Factories Act, IBR, PESO, CEA) against the inspection register: applicability × required frequency × evidence. Found 8 live statutory gaps in the demo plant.
- **Maintenance Intelligence** — per-asset MTBF, failure-mode patterns, downtime and automatic repeat-failure RCA triggers: nine identical seal failures on one pump is a process failure, not bad luck.

The flagship scenario: transformer TR-501 had a Buchholz-alarm near-miss in December 2025 because the dissolved-gas analysis on file was three months stale. In July 2026 the same 180-day statutory test is overdue again — it lives in the electrical lab's spreadsheet, and nobody has noticed. PlantBrain raises the Risk Echo automatically, citing both the incident report and the live compliance state. A prevented transformer failure is worth Rs 8–15 crore before counting the outage.

4. Architecture

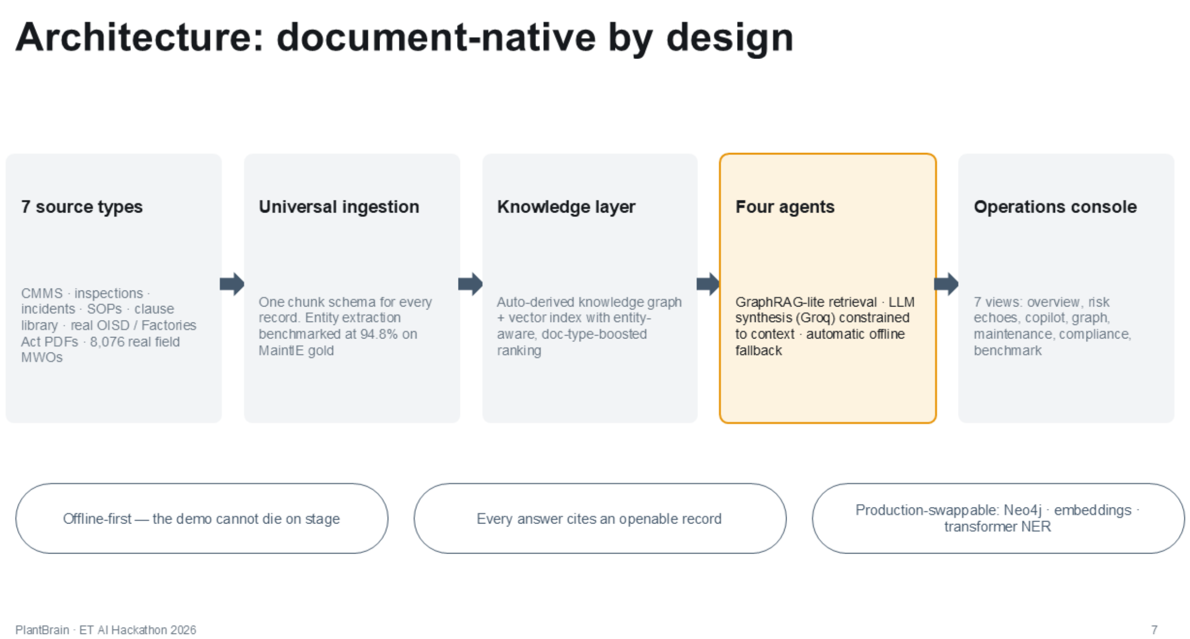

Figure 1 — end-to-end architecture of the working prototype.

- **Universal ingestion:** every record from every source — a CMMS row, a real MaintIE field work order, a page of the official OISD-STD-116 PDF, a section of an incident report — becomes one chunk schema with extracted entities (equipment tags, regulatory clause references, dates). Downstream agents never care where a fact came from.
- **Auto-derived knowledge graph:** every edge (equipment–document, clause–asset, incident–SOP) comes from entity extraction at ingest time — no manual curation, which is the property that makes adoption realistic.
- **GraphRAG-lite retrieval** (after Document GraphRAG, *Electronics* 2025; FDRKG-LLM, *Int'l J. Production Research* 2025): vector hits are expanded with 1-hop graph neighbours of the assets they mention, so structurally related documents surface with zero term overlap. Entity-aware ranking boosts records of any asset named in the query; narrative documents reach the LLM in full so a root cause is never lost below a matched summary section.
- **Offline-first:** retrieval, citations, and all four agents work with no API at all; an available LLM upgrades synthesis quality. A failed API call falls back silently. The demo cannot be broken by

connectivity.

5. Measured on Real Industrial Data

The Problem Statement 8 evaluation focus asks for entity-extraction accuracy across document types, ideally validated with real industrial document samples. We validate against **MaintIE** (Bikaun et al., LREC-COLING 2024, MIT licence): 1,076 expert double-annotated real maintenance work orders from mining, rail and heavy industry. Our extractor trains on 860 records and is scored on a held-out 216 (1,222 tokens):

Entity class	Precision	Recall	F1	Support (tokens)
PhysicalObject	0.94	0.98	0.96	475
Activity	0.95	0.94	0.95	187
Process	0.96	0.85	0.90	26
State	0.93	0.80	0.86	98
Property	1.00	0.71	0.83	7
Overall	**token accuracy 94.8%**		**macro-F1 0.909**	**1,222**

The classifier is deliberately simple (windowed lexical features → logistic regression) so the result is reproducible with one command and no API dependency; the production path swaps in a fine-tuned transformer on the same schema. Beyond the benchmark, the prototype ingests all 8,076 real MaintIE work orders and 142 pages of official regulatory documents (OISD-STD-116, OISD-STD-117 amendment, Factories Act 1948), so copilot answers about fire-protection requirements cite actual standard pages with actual design values.

6. Innovation and Differentiation

- **Risk echoes are the innovation.** Document chatbots answer questions someone thinks to ask. PlantBrain connects records nobody reads together — an incident report from December and an inspection register row from today — and speaks up unprompted.
- **Peer-reviewed architecture:** knowledge-graph-enhanced RAG outperforms plain RAG on industrial fault reasoning (Document GraphRAG, *Electronics* 2025; FDRKG-LLM, *IJPR* 2025). PlantBrain implements exactly this pattern, lightweight.
- **Category validated at billion-dollar scale:** Schneider Electric is acquiring Cognite (~$170M ARR industrial knowledge graphs); Siemens ships an Industrial Copilot reporting ~25% reactive-maintenance savings in pilots.
- **The open wedge:** those platforms are multi-year deployments priced for global majors, and none ships an Indian regulatory brain. PlantBrain is document-native — it deploys on the file shares and registers a plant already has — and compliance-first, with OISD / Factories Act / IBR / PESO / CEA built in. Land with the Compliance Radar's auditable ROI, expand to the full knowledge layer.

7. Scalability and Roadmap

Prototype (today)	Production path
TF-IDF + type/entity boosts	Embedding model + vector DB, hybrid BM25+dense
NetworkX in-memory graph	Neo4j with industrial ontology (ISO 14224)
Logistic-regression NER (benchmarked)	Transformer NER fine-tuned on the same MaintIE schema
CSV / Markdown / PDF loaders	SAP PM / Maximo connectors, OCR, email archives
Page-level PDF chunks	Layout-aware parsing + P&ID vision (cf. TCS Digitize-PID)
Streamlit console	Role-based web + mobile for field technicians

Every layer sits behind an interface chosen for swappability: the engines upgrade, the system design does not change. One deployment pattern serves steel, refining, power, cement and mining.

8. Technology Stack and How to Run

Python 3.13 · Streamlit · scikit-learn · NetworkX · Plotly · pdfplumber · Groq LLM API (llama-3.3-70b-versatile) with automatic offline fallback. Setup and run instructions are in the repository README; the benchmark regenerates with `python -m src.benchmark`.

Mapping to the stated evaluation focus

Evaluation focus item	Where PlantBrain delivers it
Entity extraction accuracy across document types	94.8% / 0.909 macro-F1 on MaintIE gold; per-class table in-app
Query answer quality on domain benchmark questions	Copilot with cited, context-constrained answers over 8,540 records

Knowledge graph linkage completeness	54-node auto-derived graph; graph-sourced citations badged in every answer
Time-to-answer versus traditional search	Seconds versus the 35%-of-hours baseline; live in the demo
Compliance gap detection accuracy	8 planted statutory gaps found (OISD/CEA/Factories Act), zero false compliant
Validation with real industrial document samples	8,076 real MWOs + official OISD-STD-116 + Factories Act 1948 ingested

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Data notes: the demo plant (Bharat Ispat & Energy Ltd, Unit 2) is fictional and generated; the MaintIE corpus is MIT-licensed; OISD documents were obtained from oisd.gov.in and are used for local demonstration only, not redistributed. All third-party sources are credited in the repository.